BIG MOUNTAIN RESORT

$81.00	$95.87	+$14.87	~$0.88
Current Ticket Price	Model-Supported Price	Pricing Opportunity	New Chairlift Cost/Ticket

1. Problem Statement

Big Mountain Resort (Montana, 105 trails) recently installed a new chairlift adding **$1,540,000** in seasonal operating costs. The business needed a data-driven answer to two questions: are we pricing our tickets correctly, and which facility investments will strengthen our market position?

A machine learning model was trained on competitor US resort data to predict the ticket price Big Mountain's facilities can support — and four investment scenarios were modelled. Revenue calculations assume **350,000 annual visitors** buying an average of **5 tickets each**.

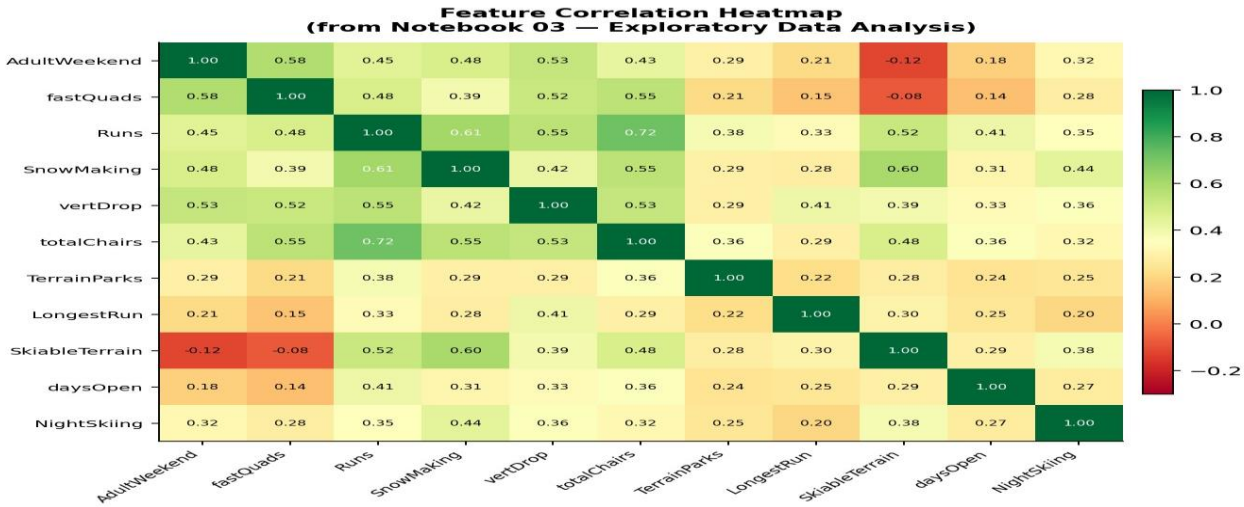
2. Data Wrangling and Exploratory Analysis

The raw dataset had **330 US ski resorts and 27 features.** Key cleaning steps:

- **fastEight column dropped** — over 50% missing, all remaining values were zero.
- **Silverton Mountain corrected** — skiable area recorded as 26,819 acres (transcription error); confirmed real value is 1,819 acres.
- **53 rows dropped** — resorts with no ticket price data at all. Final dataset: **277 rows, 25 columns.**
- **Target: AdultWeekend** — Montana resorts charge identical weekday/weekend prices; weekend had fewer missing values.

State PCA captured **77% of variance in 2 components** with no geographic pricing pattern — all states treated equally.
The correlation heatmap below confirms fast quads, vertical drop, and snow making as the strongest positive predictors.
Skiable terrain area is slightly negative — more terrain with the same lifts means thinner coverage per skier.

Figure 1: Feature Correlation Heatmap (Notebook 03 — EDA, Section 3.5.



5.2).

Green = strong positive correlation with ticket price. fastQuads, vertical_drop, Snow Making_ac, and Runs are the top drivers. SkiableTerrain_ac is slightly negative.

3. Modelling and Evaluation

Preprocessing	Median imputation, StandardScaler (linear only), SelectKBest k=8 via GridSearchCV
Train / Test	70% train, 30% test — random_state=47 for reproducibility
Validation	5-fold cross-validation on all models before final test evaluation
Big Mountain	Excluded from training — price predicted purely from competitor data

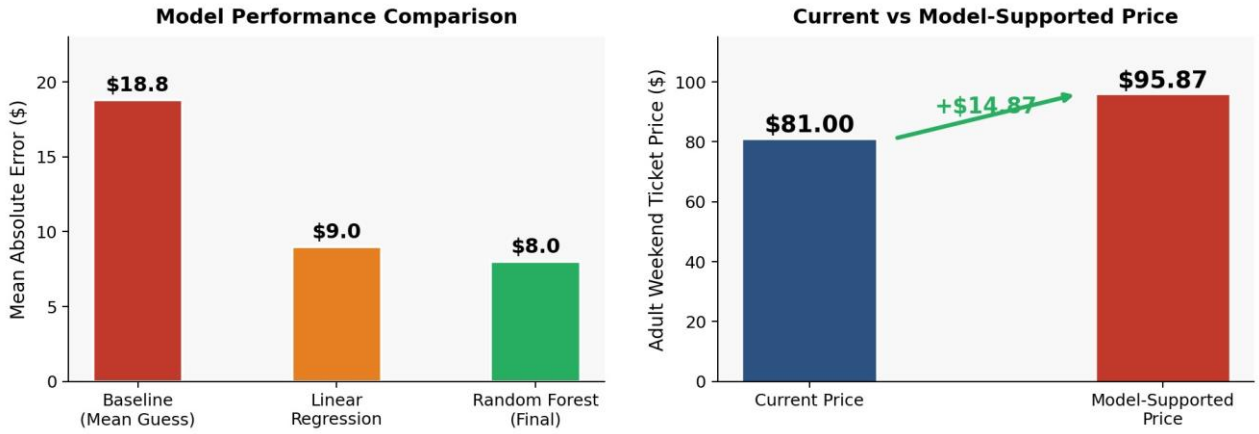


Figure 2 (left): Model MAE comparison — Random Forest cuts the baseline error by over 57%. Figure 2 (right): Big Mountain's current $81.00 vs the model-supported $95.87 — a $14.87 pricing gap.

- **Baseline (Dummy):** Always predicts the training mean. MAE = $18.80, $R^2 = 0$. The floor every real model must beat.
- **Linear Regression:** Median imputation + scaling + SelectKBest (k=8). Explains 80%+ variance on train, 70%+ on test. MAE ≈ $9.00. CV and test consistent.
- **Random Forest (selected):** Median imputation, no scaling needed. CV MAE ≈ $8.00 — nearly $1 lower than linear regression, less fold variability, test matched CV.

Why Random Forest? Better accuracy, handles non-linear patterns naturally, stable cross-validation, and test results matched CV — clear signs the model genuinely generalises to unseen data.

4. Pricing Recommendation

Ticket-days/season	350,000 visitors x 5 = 1,750,000
New lift cost/ticket	$1,540,000 / 1,750,000 = ~$0.88
Pricing gap	$95.87 - $81.00 = $14.87 (17x the lift cost per ticket)
Break-even	A $1/ticket increase = $1,750,000 — covers the lift cost in full

How to present to leadership: Lead with the facility comparison charts — once executives see Big Mountain in the top quartile on snow making, chairs, vertical drop, fast quads, and runs while charging less than inferior competitors, the $95.87 lands as a natural confirmation. Recommend a phased approach: $86 in Year 1, $91 in Year 2. Be honest about the ~$8 model margin of error.

5. Scenario Analysis

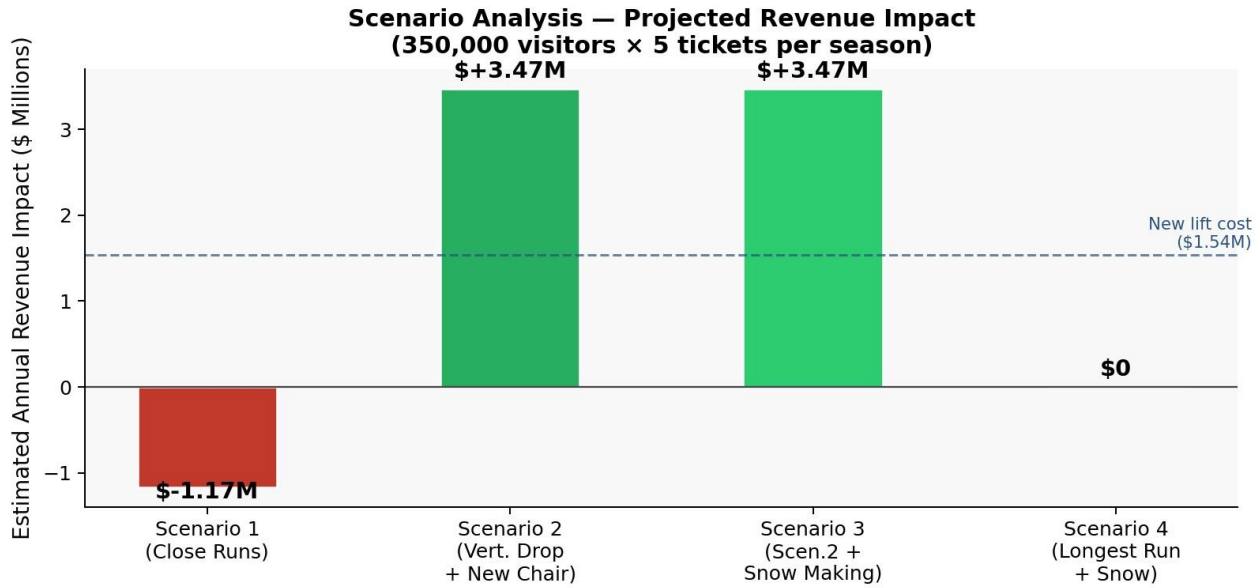


Figure 3: Projected annual revenue impact by scenario (350k visitors x 5 tickets). Scenario 2 generates +$3.47M — more than double the new lift's $1.54M operating cost (dashed blue line).

- **Scenario 1 — Close Runs:** Only reduces revenue. Closing 1 run = no impact. Closing 2-5 = gradual drop (~$0.67/ticket). Closing 6+ = large drop (>$1.26/ticket). Pilot 1-2 runs off-peak before any permanent decision.
- **Scenario 2 — Vertical Drop +150ft + New Chair RECOMMENDED:** +$1.99/ticket, estimated +$3,474,638 seasonal revenue. Targets the two strongest price drivers simultaneously.
- **Scenario 3 — Scenario 2 + 2 Acres Snow Making:** Identical result to Scenario 2. Big Mountain already leads the market on snow coverage — 2 extra acres adds nothing to price.
- **Scenario 4 — Extend Longest Run + Snow Making NOT RECOMMENDED:** Zero revenue impact. Longest run ranks very low in feature importance.

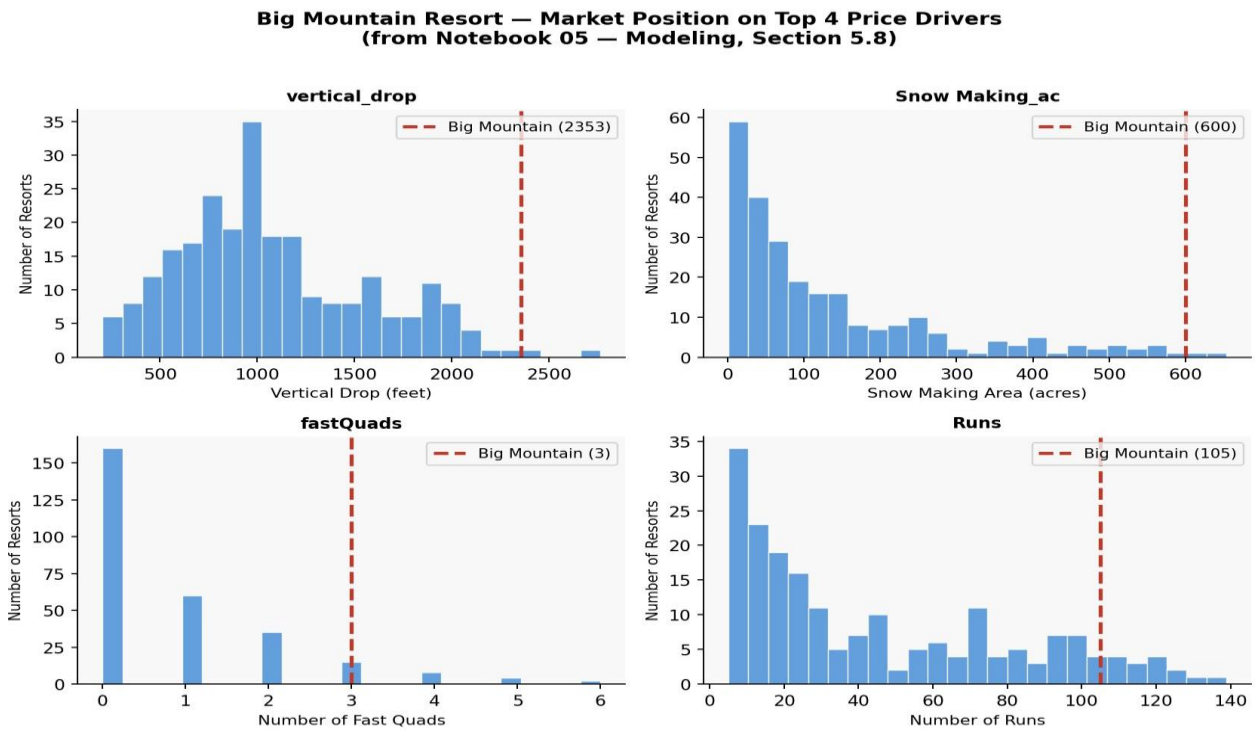


Figure 4: Big Mountain's position vs all US resorts on the 4 key price drivers (Notebook 05 — Modeling, Section 5.8). The red dashed line marks Big Mountain — well above the market average on every metric, confirming the $14.87 underpricing gap.

6. Conclusion and Future Scope

Conclusion

Big Mountain has elite-level facilities yet charges less than resorts with fewer amenities. The model confirms a **$14.87 pricing gap.** The new chairlift at $0.88/ticket is easily absorbed within this headroom. **Scenario 2** (vertical drop extension + new chair) is the priority investment at an estimated +$3.47M seasonal revenue. Raise prices in phases — $86 then $91 — while piloting run closures carefully.

Future Scope

- Real visitor count data to replace the assumed 350k figure — improves revenue accuracy.
- Full operating cost breakdown (staffing, maintenance, utilities) to assess profitability, not just revenue.
- Demand elasticity analysis — understand how visitor numbers respond to price increases.
- Deploy as a Streamlit self-service tool so analysts can run scenarios without needing a data scientist each time.
- Incorporate multi-year data and guest satisfaction scores for a richer model.